Olga Balsalobre-Ruza

PhD candidate [defending in May 2026]

Curriculum Vitæ

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See personal webpage
Spanish [Native] & English [Advanced, C1]

I hold a degree in Physical Sciences with a major in Astrophysics from the Complutense University of Madrid (UCM). During my studies, I did internships across diverse areas of astrophysics that helped me build both technical skills and scientific independence, including high-energy astrophysics, observational astronomy, and exoplanetary science.

Currently, I am a fourth-year PhD student at the Center for Astrobiology (CAB) in Madrid, specializing in the **search and characterization of exoplanets and co-orbital systems** using multiple techniques, with a strong focus on **radial velocities** (RV) and **direct imaging** (DI). I have designed observational strategies for RV surveys and gained extensive experience with both ground- and space-based observatories, being the **PI of 8 approved observing programs** (195 h telescope time granted), and co-I of 13 programs (3974 h). I am an enthusiastic Python programmer with a keen interest in expanding my skills, for instance in machine learning and astro-statistics.

During the PhD, I have published **4 first-author** and **7 co-author papers**, developed **2 Python packages**, and presented my research at international conferences and institutions (**17 talks** and **6 posters**). I have also communicated my work to the public through several media. I thrive in both independent and collaborative research environments, valuing teamwork as essential for scientific progress.

Keywords: Exoplanetary Systems; Planetary Formation; Detection Techniques; Data Analysis; Bayesian Statistics

1 Education

Bachelor in Physics
UCM
- Honors in three subjects
- Speciality in Astrophysics

BSc thesis
Gamma Rays data analysis detected with Cherenkov telescopes (CTA)

2015 - 2020

Master in Astrophysics
UCM
- Honors in two subjects
- Honors in the MSc thesis

MSc thesis
Development of `KOBEsim` [see coding Section]

2020 - 2021

PhD in Astrophysics
UCM & CAB
Supervised by Dr. Jorge Lillo-Box & Dr. Nuria Huélamo

PhD thesis
Search for co-orbital exoplanets via RV, DI & transits (T)

2022 - [May 2026]

2 Publications in peer-reviewed journals

The observational technique used is shown after the hyphen as radial velocity **(RV)**, direct imaging **(DI)** and transits **(T)**.

First author

(5) **Balsalobre-Ruza** et al., in prep. – **DI**
Inner dust clumps in PDS 70 revealed by SPHERE with star-hopping

(4) **Balsalobre-Ruza** et al., 2025 [A&A, 694, A15] – **RV** & **T**
KOBE-1: The first planetary system from the KOBE survey. Two planets likely in the sub-Neptune mass regime around a late K dwarf

(3) **Balsalobre-Ruza** et al., 2024 [A&A, 689, A53] – **RV** & **T**
The $\mathcal{TROY}$ project: III. Exploring co-orbitals around low-mass stars

(2) **Balsalobre-Ruza** et al., 2023 [A&A, 675, A172] [ESO press release] – **DI**
Tentative co-orbital submillimeter emission within the Lagrangian region L5 of the protoplanet PDS 70 b

(1) **Balsalobre-Ruza** et al., 2022 [A&A, 669, A18] – **RV**
`KOBEsim`*: A Bayesian observing strategy algorithm for planet detection in radial velocity blind-search surveys*

Co-author

(7) Grouffal et al., 2025 [Accepted in A&A, arXiv:2507.01807] – **RV**
The star HIP 41378 potentially misaligned with its cohort of long-period planets
Contribution: Analysis and text on CARMENES RV, Sect. 2 (author 9/32)

(6) Figueira et al., 2025 [A&A, 700, A174] – **RV**
A comprehensive study on radial velocity signals using ESPRESSO: Pushing precision to the 10 cm/s level
Contribution: Scientific feedback on photometric centroid analysis (author 9/33)

(5) Cesario et al., 2024 [A&A, 692, A172] — **DI**
Large Interferometer For Exoplanets (LIFE). XIV. Finding terrestrial protoplanets in the galactic neighborhood
Contribution: Manuscript revision (author 27/64)

(4) Lillo-Box et al., 2024 [A&A, 689, L8] — **RV & T**
K2-399 b is not a planet: The Saturn that wandered through the Neptune desert is actually a hierarchical eclipsing binary
Contribution: Manuscript revision (author 8/22)

(3) Lillo-Box et al., 2024 [A&A, 686, A232] — **DI**
The AstraLux-TESS high spatial resolution imaging survey. Search for stellar companions of 215 planet candidates from TESS
Contribution: Scientific feedback on photometric centroid analysis (author 4/9)

(2) Christiaens et al., 2023 [ASCL redord:2311.002] — **DI**
`VCAL-SPHERE`*: Hybrid pipeline for reduction of VLT/SPHERE data*
Contribution: Minor coding contribution (author 5/5)

(1) Lillo-Box et al., 2022 [A&A, 667, A102] — **RV**
The KOBE experiment: K-dwarfs Orbited By habitable Exoplanets. Project goals, target selection, and stellar characterization
Contribution: Sample selection and Sect. 4.3 (author 8/36)

Book Chapters

(1) Lillo-Box & **Balsalobre-Ruza** (to be published), Elsevier
Exoplanet Detection Methods. Chapter 8. Orbital Phase Modulations

3 Research experience

- **Predoctoral Researcher [PhD thesis]** — CAB (Madrid, Spain)
  Search for co-orbital exoplanets in the $\mathcal{TROY}$ project — Mar 2022 – Present [3 years]
  - **Predoctoral stay** at Liège University (Belgium)
    Collaboration with Dr. Valentin Christiaens on high-contrast imaging — March/August 2025 [3 weeks]
  - **KOBE hackathons** — CAB (Madrid), LAM (Marseille), GO (Genève)
    Three separate hackathons for RV analysis with the KOBE team — May 2024, May & Nov 2025 [3 weeks]
  - **Predoctoral stay** at University California Santa Cruz (USA)
    Training in N-body simulations and collaboration with Dr. Nathaly Bathala's group — Jun – Aug 2024 [2 months]
  - **Predoctoral stay** at ESO Santiago (Chile)
    Training in the analysis of ALMA observations with Dr. Itziar de Gregorio — Oct 2022 – Nov 2022 [1 month]
- **Research Assistant** — Universidad Autónoma de Madrid (UAM) (Spain)
  Project on exoplanet dynamics — Dec 2021 – Mar 2022 [4 months]
- **Summer Internship** — CEFCA (Teruel, Spain)
  Characterization of atmospheric extinction at the Javalambre Observatory — Jul 2021 – Sep 2021 [3 months]
- **Internship Trainee** — CAB (Madrid, Spain)
  Support for Mars data analysis with the Rover Curiosity — Feb 2021 – Jun 2021 [5 months]
- **Internship Trainee** — CIEMAT (Madrid, Spain)
  Characterization of the LST-1 Cherenkov telescope camera — Feb 2019 – Feb 2020 [1 year]

4 Coding

Python

- **6-year** experience as daily user
- **54-hour** online courses
- **Codes** available in GitHub:
  - `moca` — Developer of an RV simulator of planetary signals including activity with Gaussian processes. Focused on preparing observing proposals and injection recovery for sensitivity analysis [to be published]
  - `VCAL-SPHERE` — Collaborator in a reduction, pre-processing and post-processing SPHERE pipeline
  - `KOBEsim` — Developer of a Bayesian algorithm to plan RV observations to boost the detection of exoplanets

Machine Learning & AI

- **240-hour** course by Samsung Innovation in the Polytechnic University of Madrid (UPM) in machine and deep learning
- **18-hour** course organized by IPARCOS group at UCM applied to physics and astrophysics
- Application of Random Forest algorithms in my **BSc thesis** with Python

Additional experience

- **Languages:** `Matlab`, `R`, `IDL`, `Stan`, `SQL`, `HTML`
- **SO:** macOS, Linux and Windows
- **Some astrophysics-relevant packages/programs:**
  - **Visualization:** `carta`, `DS9`
  - **Radial velocity:** `radvel`, `kima`, `serval`, `raccoon`, `KOBEsim`, `moca`
  - **Direct imaging:** `VCAL`, `VIP`, `star-hopping`, `CASA`, `irdap`, `LIFEsim`
  - **Transits:** `lightkurve`, `wotan`, `batman`
  - **General:** `astropy`, `emcee`, `bayev`, `george`, `rebound`, `keras`

5 Observational experience

Visitor mode observations

- **HARPS-N [6 nights]** in the TNG (La Palma Observatory) [Jun 2024]
- **ESPRESSO [1 night]** in the VLT (ESO, Paranal Observatory) [Feb 2024]
- **HARPS [6 nights]** in the 3.6 m (ESO, La Silla Observatory) [Aug 2022]

Accepted observing programs

- **PI programs [195 h granted]**:
  (8) **SPHERE [2 h]** (ESO, Paranal Observatory) [P116]
  (7) **GRAVITY [4 h]** & **SPHERE [2 h]** (ESO, Paranal Observatory) **[DDT P115]**
  (6) **CARMENES [39 h]** (CAHA Observatory) [25A]
  (5) **CHEOPS [66 h]** (ESA) [AO-5]
  (4) **CARMENES [40 h]** (CAHA Observatory) [24B]
  (3) **SPHERE [4.5 h]** coronagraphic IRDIS DPI star-hopping (ESO, Paranal Observatory) [P113]
  (2) **CARMENES [11 h]** (CAHA Observatory) [23B]
  (1) **CARMENES [22 h]** (CAHA Observatory) [22B]
- **Co-I programs [3860 h granted]**:
  (13) **ESPRESSO [57 h]** (ESO, Paranal Observatory) – PI: A.Súarez-Mascareño [P116]
  (12) **CARMENES [51 h]** (CAHA Observatory) – PI: I.Mendigutia [25B]
  (11) **CARMENES [20 h]** (CAHA Observatory) – PI: C. Cifuentes [25B]
  (10) **CARMENES [10 h]** (Calar Alto Observatory) – PI: J. Lillo-Box [25A]
  (9) **CAFE [70 h]** (Calar Alto Observatory) – PI: J. Lillo-Box [25A]
  (8) **CAFE [90 h]** (Calar Alto Observatory) – PI: J. Lillo-Box [24B]
  (7) **CHEOPS [32 h]** (ESA) – PI: Jorge Lillo-Box [AO-5]
  (6) **CARMENES [20 h]** (CAHA Observatory) – PI: J. Lillo-Box [24A]
  (5) **HARPS-N [54 h]** (La Palma Observatory) – PI: J. Lillo-Box [24A]
  (4) **NIRPS & HARPS [4 h]** (ESO, La Silla Observatory) – PI: J. Lillo-Box [P112]
  (3) **ESPRESSO [40 h]** (ESO, Paranal Observatory) – PI: J.P. Faria [P112]
  (2) **CAFOS [20 h]** (CAHA Observatory) – PI: A. Castro-González [22B]
  (1) **CARMENES [3500 h]** (CAHA Observatory) The KOBE experiment (legacy program) – PI: J. Lillo-Box [21A–26B]

6 Dissemination

Seminars

(6) Geneva Observatory — Genève (Switzerland), Apr 2025
The TROY project: Do co-orbital exoplanets really exist?

(5) Liège University — Liège (Belgium), Mar 2025
The TROY project: Do co-orbital exoplanets really exist?

(4) ESA exoplanet team — Madrid (Spain), Oct 2024
The TROY project: Do co-orbital exoplanets really exist?

(3) ESO headquarters — Santiago (Chile), Nov 2022
`KOBEsim`. *Bayesian observing strategy algorithm for planet detection in RV surveys*

(2) ESO headquarters — Santiago (Chile), Aug 2022
PiCs. Planets in Giant stars

(1) CEFCA — Teruel (Spain), Sep 2021
Javalambre Observatory atmospheric extintion

Conferences

- **Talks**

(11) Madrid-Area Exoplanet Science Meeting at ESAC — Madrid (Spain), Oct 2025
Looking closer at PDS 70: Tracing Trojan dust?

(10) Detection and Dynamics of Exoplanets — Coimbra (Portugal), Jul 2025
Looking closer at PDS 70: Tracing Trojan dust?

(9) Madrid-Area Exoplanet Science Meeting at ESAC — Madrid (Spain), Dec 2024
The KOBE experiment. First results

(8) Bay Area Exoplanets Meeting — Santa Cruz, CA (USA), Jul 2024
The TROY project: co-orbital exoplanets and how to find them

(7) Other Worlds Laboratory Summer Program 2024 — Santa Cruz, CA (USA), Jul 2024
Developing `Moca` *to compute mock RVs induced by exoplanets*

(6) Keep uPhDated CAB — Madrid (Spain), Dec 2023
Genesis of Trojans in PDS 70 b potentially hunted with ALMA

(5) Toward Other Earths III — Porto (Portugal), Jul 2023
Genesis of Trojans in PDS 70 b potentially hunted with ALMA

(4) PhD meetings UCM — Madrid (Spain), Mar 2023
Exotrojan planets. Missing pieces of planetary formation

(3) Keep uPhDated CAB — Madrid (Spain), Dec 2022
Exotrojan planets. Missing pieces of planetary formation

(2) Madrid-Area Exoplanet Science Meeting at ESAC — Madrid (Spain), Oct 2022
In the search for exotrojan planets

(1) Spanish Society of Astronomy XV at IAC — Tenerife (Spain), Sep 2022
The KOBE experiment - `KOBEsim`*: improving RV detection through efficient scheduling*

- **Posters**

(6) Extremely Precise Radial Velocities 6 (EPRV 6) — Porto (Portugal), Jun 2025
Exotrojans around low-mass stars: A search with a RV clock

(5) Extremely Precise Radial Velocities 6 (EPRV 6) — Porto (Portugal), Jun 2025
KOBE-1: A new multi-planetary system around a late K dwarf

(4) Complex Planetary Systems II — Namur (Belgium), Jul 2023
Witnessing the genesis of exotrojans in PDS 70 [Awarded]

(3) ESLAB 2023 ESTEC [Remote participation] — Noordwijk (Netherlands), Mar 2023
Towards completing extrasolar systems with the TROY project

(2) Scientific Meeting of the Spanish Society of Astronomy XV — Tenerife (Spain)
The KOBE experiment - `KOBEsim`: improving RV detection through efficient scheduling — Sep 2022

(1) LATSIS 22 ETH [Remote participation] — Zurich (Switzerland)
Co-orbital worlds as possible bullets of giant impacts — Sep 2022

Technical talks

- *PyCoffee* — Python short seminars series — CAB, Madrid
  Regular expressions for a more efficient work — Jan 2025
  Start doing your own Python modules — Mar 2024
- *Journal Club* — Paper presentation series — CAB, Madrid
  Reviewing: DSHARP VII. The Planet-Disk Interactions Interpretation — Nov 2024

Outreach and media

- Nacional Radio and international written interviews on my research
  Onda Regional de Murcia, Radio Extremadura, ABC, Science, Space among others
- Press-releases on my research work
  ESO, CAB, CAHA
- Collaborator in the outreach YouTube channel of my research group
  Remote Worlds Lab
- First-authored article for the Revista Española de Física
  Acercándonos al desierto donde residen los mundos habitables, V39, N2 (2025)
- Four outreach talks in schools
  Hidden planets in the immense Universe (Madrid, Apr 2025) Le dijo un Quark al Cosmos (Jun 2020 - Jan 2022) [Not available anymore]
- Volunteer in the ESAC Open Day 2023 (Madrid)

Courses on communication skills

- Course on TV, Radio and News Conferences for researches
  Organized by the Spanish National Radio and television (RTVE) and CSIC (17 h, Feb 2025)
- Writing better proposals and Papers workshop
  Organized by Dr. Artie Hatzes
- Speak in public and present work efficiently course
  Organized by UCM

7 Other event attendance

- *exoELT: Planetary Formation and Exoplanets in the ELT Era* — ESO Garching headquarters (Germany), Nov 2025
- *PLATO week #16* — Paris (France), Oct 2025
- *General seminars at CAB* — online, 2024–2025
- *Symposium (IAUS 387): (Toward) Discovery of Life Beyond Earth and its impact* — online, Apr 2024
- *Accessing the Gaia treasure trove with TOPCAT, STILTS, ADQL and Gaia Sky* — ESO (Chile), Nov 2022

8 Organizing experience

- **LOC** of a workshop in ESAC — Madrid (Spain)
  Exploring Tatooine and beyond: Circumbinary planets with ESA missions — Dec 2025
- **Co-organizer** of PyCoffee series at CAB — Madrid (Spain)
  See Github repository — Nov 2023 – Sep 2025
- **SOC** of a mini-symposium within the XV meeting of the Spanish Society of Astronomy — Granada (Spain)
  Here and There: Synergies between Solar System and Exoplanetary Research — Jul 2024

9 Teams and Consortiums

- The TROY project | `https://www.troy-project.com/#team`
- The KOBE experiment | `https://kobe.caha.es/#team`
- PLATO Consortium

10 Funding

- *Time Evolution and Metamorfosis of exoPlanets and their astmOspheres (TEMPO)*, PI: J. Lillo-Box 200 000 €
  Research Spanish Agency CNS2023-144309

- *Stellar and Planetary Formation: Diversity and Habitability using JWST and PLATO*, PI: D. Barrado-Navascués 260 000 €
  Research Spanish Agency PID2023-150468NB-I00